

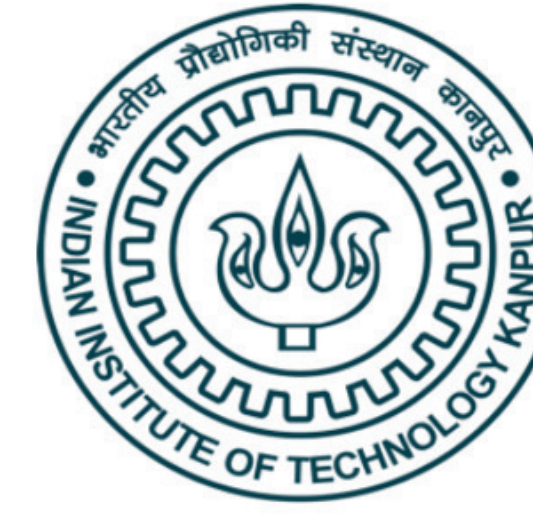


Recovery of Active Cathode Material from Spent Mobile Lithium-ion Batteries using a Custom-made Ternary Deep Eutectic Solvent

K Appurva¹, Rajesha P M¹, P Hemanth², V M Aravind², Raju K Gupta^{2*}, Sudhir H Ranganath^{1*}

¹Department of Chemical Engineering, Siddaganga Institute of Technology, Tumakuru *Corresponding author: sudhirh@sit.ac.in

²Department of Chemical Engineering, Indian Institute of Technology, Kanpur *Corresponding author: guptark@iitk.ac.in



Abstract

Lithium-ion batteries (LiBs) are integral to modern technology, powering a broad array of devices, including mobile phones, laptops, electric vehicles, and renewable energy systems. However, their widespread utilization presents considerable environmental and social challenges, due to the reliance on scarce and hazardous materials that necessitate energy-intensive extraction. Recycling of LiBs emerges as a viable strategy to address these issues, offering a method to reclaim valuable metals and reduce the demand for the extraction of fresh raw materials. It initiates the potential to generate economic opportunities and foster innovation within the battery sector. Conventional recycling methods, including pyrometallurgical and hydrometallurgical processes, are energy-intensive and hazardous. To address this, we developed a novel Ternary Deep Eutectic Solvent (TDES), which is green and composed of Hydrogen Bond Donors and Acceptors in a specific molar ratio. This solvent offers several advantages, such as low freezing points, high thermal stability, low volatility, and tunability. Spent mobile phone batteries were dismantled under water to safely recover Active Cathode Material (ACM), which contained Li, Co, Ni, and Mn. The TDES achieved rapid leaching of the ACM at 90 °C within 30 minutes and at a significantly lower operating temperature (90 °C) than traditional methods (~700 °C). High extraction efficiencies of ~99% were achieved for Li⁺, Co²⁺, Ni²⁺, and Mn²⁺. Subsequent stage-wise selective precipitation allowed recovery of high-purity metal oxides of Nickel, Manganese and Cobalt and Lithium is selectively precipitated as Lithium carbonate. Kinetic analysis indicated that leaching was controlled by both Liquid-film diffusion and Product-layer diffusion for Li and Co. The same TDES was applied to LiNi_{0.85}Co_{0.1}Al_{0.05}O₂ (NCA), yielding extraction efficiencies of ~68% Ni, ~64% Co, ~92% Al, and ~70% Li, aligning with elemental analysis. Further optimization of metal recovery from NCA systems is under current investigation. Overall, the developed TDES demonstrates strong potential as a sustainable and efficient medium for LiB recycling.

Introduction

- Lithium, “An important metal to promote the world”, is widely utilized in various fields including batteries, aerospace, nuclear power generation, and pharmaceutical industry
- Lithium sources in nature: Sedimentary rocks, Sea Water (0.1-0.6 ppm), Geothermal water (100-250 ppm), and Volcanic rocks (100-5990 ppm), Quartz (0.17-100 ppm)
- The global LiB market is expected to grow at a CAGR of 13.5% by 2029, leading to a rise in End of Life (EOL) batteries ~1 million battery packs by 2030 (Fig.1) and ~1.9 million by 2040

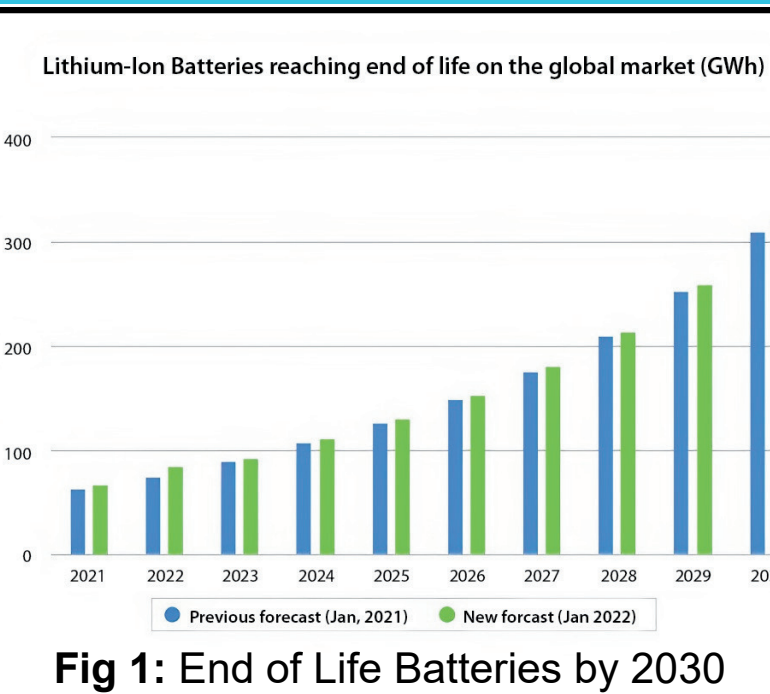


Fig 1: End of Life Batteries by 2030

Limitations of Conventional Methods of Recycling LiBs

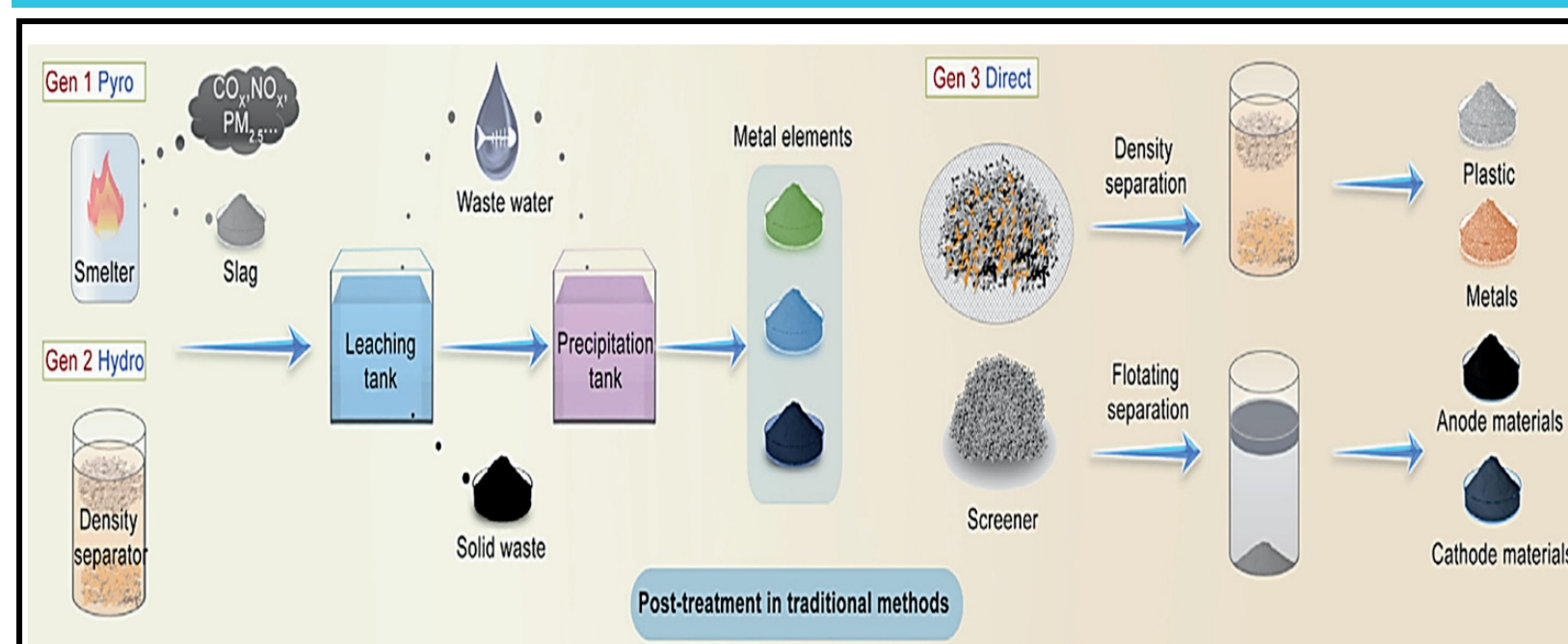


Fig 2: Post treatment in traditional LiB recycling methods

- Pyrometallurgical recovery of metals from spent LiBs is energy-intensive, low material recovery and hazardous due to release of toxic gases
- Dimethyl Sulfoxide (DMSO), N-Methyl-2-Pyrrolidone (NMP), Dimethylacetamide (DMAC), and inorganic mineral acids - currently used as solvents for metal extraction, but they pose adverse environmental impact
- Therefore, promoting effective and greener solvents for leaching is essential to mitigate the hazards posed by organic solvents while selectively and efficiently extracting valuable elements, including Lithium, from used LiBs is required

Conventional LiB Recycling Technologies

- Gen 1: Pyrometallurgical process
- Gen 2: Hydrometallurgical process
- Gen 3: Physical processes to separate cathode and anode materials

Valuable metals such as Ni, Co, Mn, and Li are mainly recovered from the cathode, the most expensive component of LiBs, contributing ~1/3 of the total battery cost.

Hypothesis: Deep Eutectic Solvents (DESs)

- Deep Eutectic Solvents (DESs) are prepared as a mixture of a Hydrogen Bond Donor (HBD) and a Hydrogen Bond Acceptor (HBA) in an appropriate ratio, which forms a liquid at room temperature
- The term “deep” in DES implies that the mixture forms a liquid with a **dramatic decrease in the melting point** because of intermolecular Hydrogen bonding between the HBD and HBA
- DESs are characterized by negative deviations from the ideal behaviour for their interactions

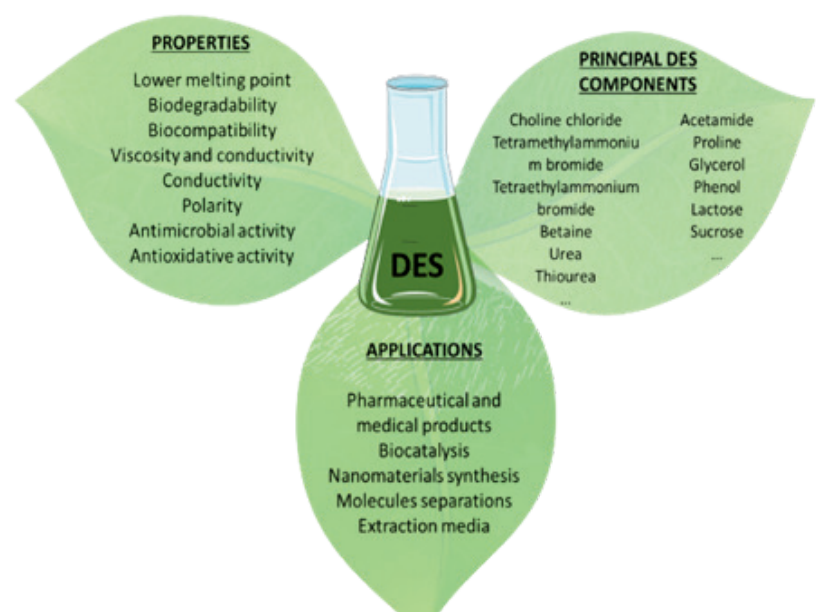


Fig 3: Characteristics of DESs

Properties:

- Tunability and Tailoring
- Higher selectivity
- Low volatility and flammability
- Biodegradability
- Wide solubility range
- Good thermal stability

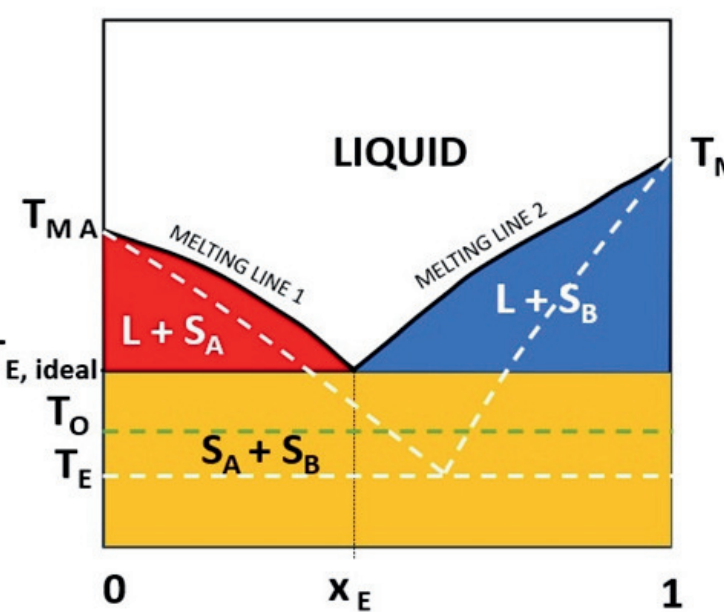


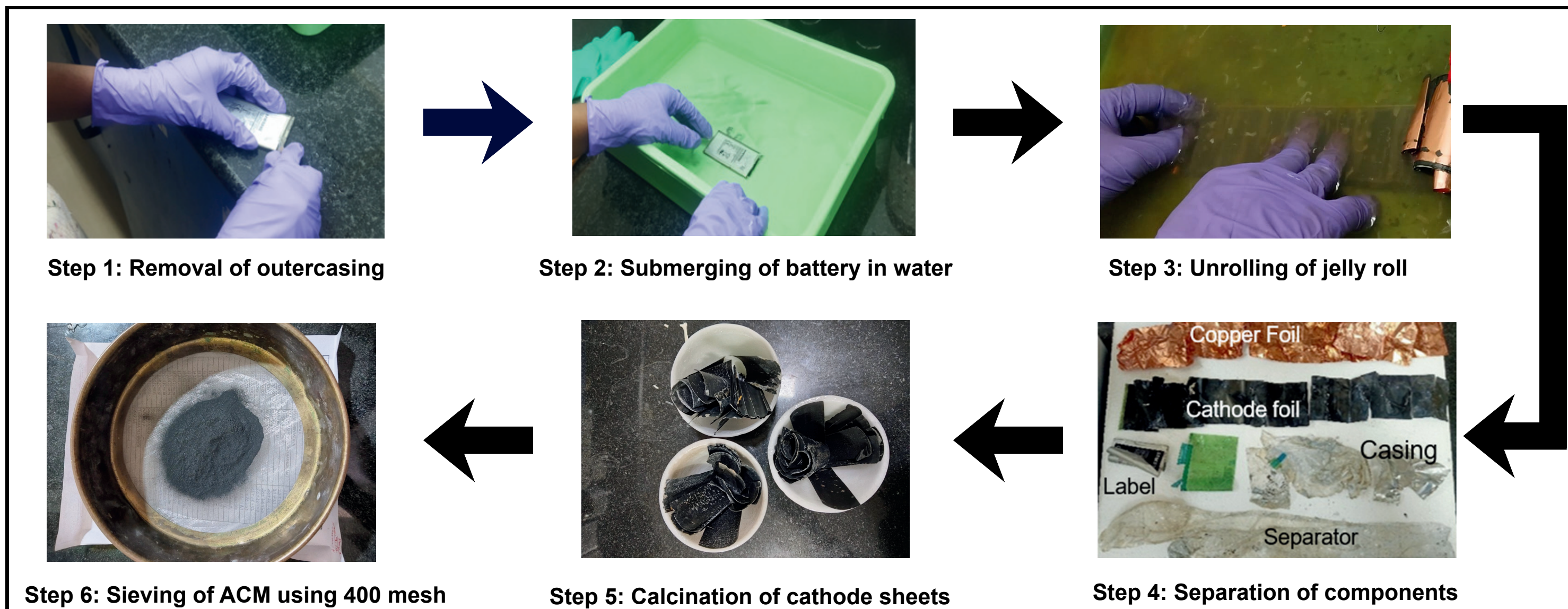
Fig 4: Melting point depression of DES

- Ternary Deep Eutectic Solvents (TDES) are an advanced class of DES that incorporate a third component, offering enhanced properties over conventional DES
- TDES offers a **greater depression in melting point** and can be easily stored and used for extraction at room temperature or lower temperatures
- The addition of a third component allows for fine-tuning of physicochemical properties such as viscosity, polarity, and solvation ability. **TDES can consists of a single HBA and two HBDs or two HBAs and a single HBD**

Objectives

- To synthesize a custom-made TDES in different molar ratios
- To carry out leaching of ACM with TDES and to selectively precipitate metal oxalates and carbonates from the supernatant obtained post-leaching
- To study the mechanisms of mass transfer and reaction on leaching kinetics

LiB Dismantling Process



Synthesis and Characterization of TDES

- For the synthesis of DES, components A, B which are HBAs and C which is an HBD, were mixed in 1:1:1, 1:1:2, 1:2:1 and 2:1:1 molar ratios.

Characterization Techniques

- FT-IR: To study functional groups in DES
- DSC: To check the thermal stability of DES
- Viscometer: To measure temperature-dependent viscosity
- NMR: To confirm H-bonding and chemical purity

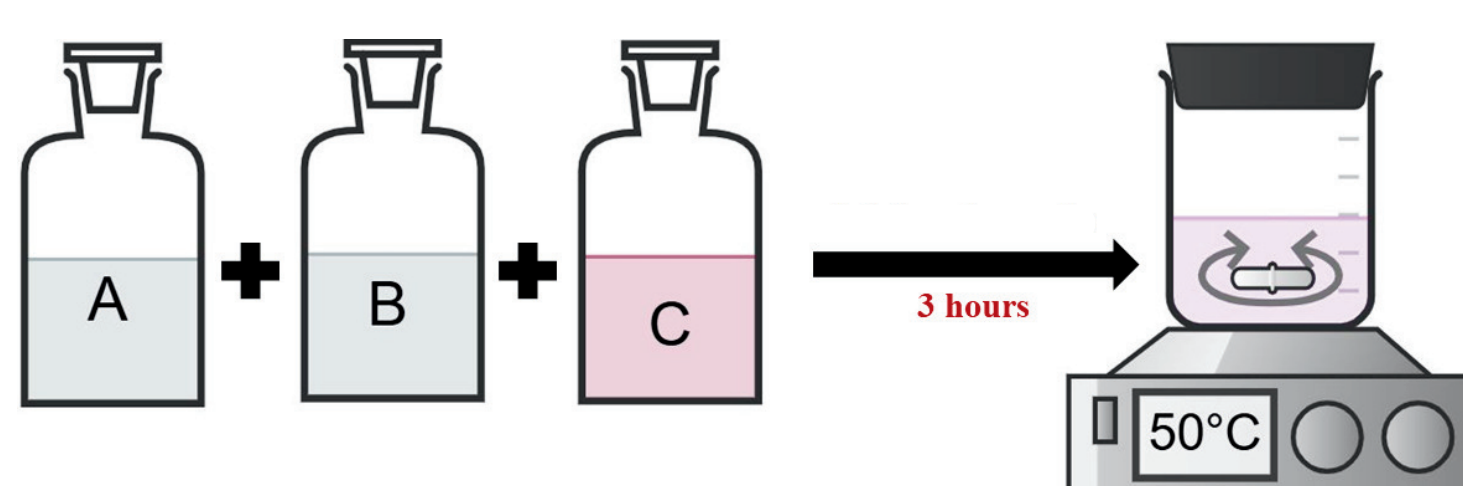


Fig 5: Synthesis of TDES using Heating and stirring method

- At higher temperatures, esterification reactions may occur. To avoid this, the synthesis of TDES was carried out at a lower temperature (50 °C)

Metal Leaching (Recovery) Process

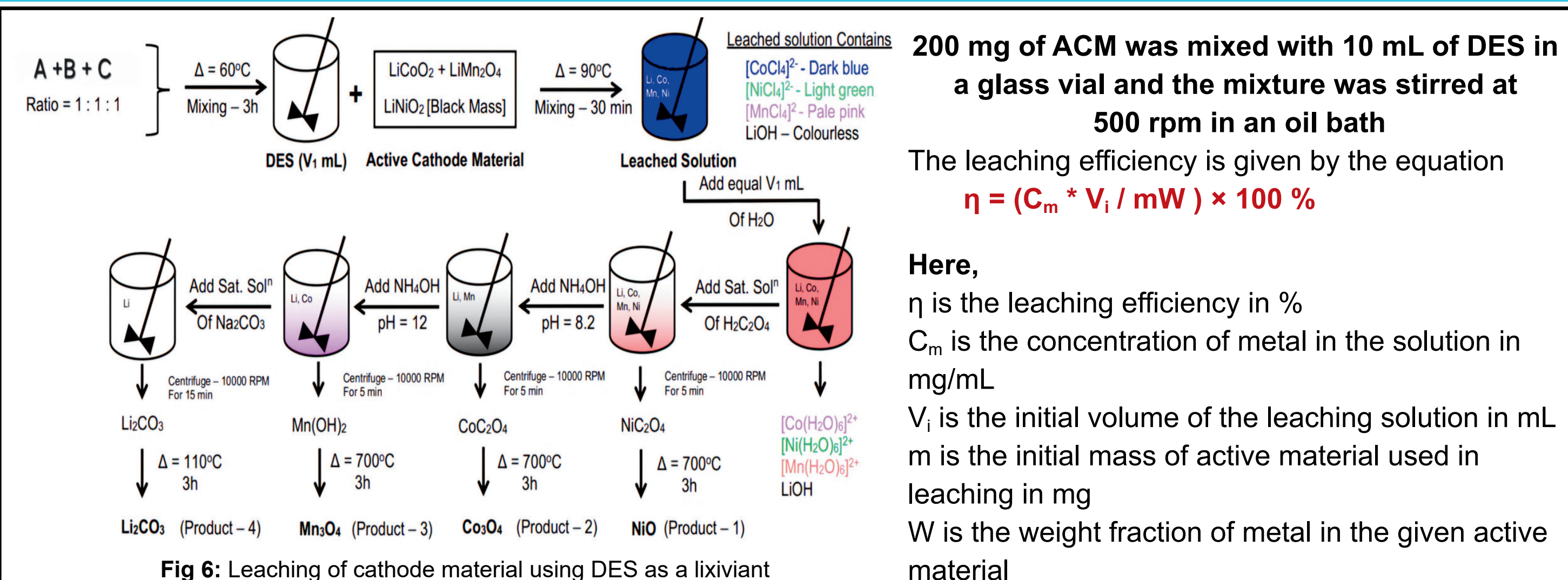


Fig 6: Leaching of cathode material using DES as a lixiviant

Results and Discussion

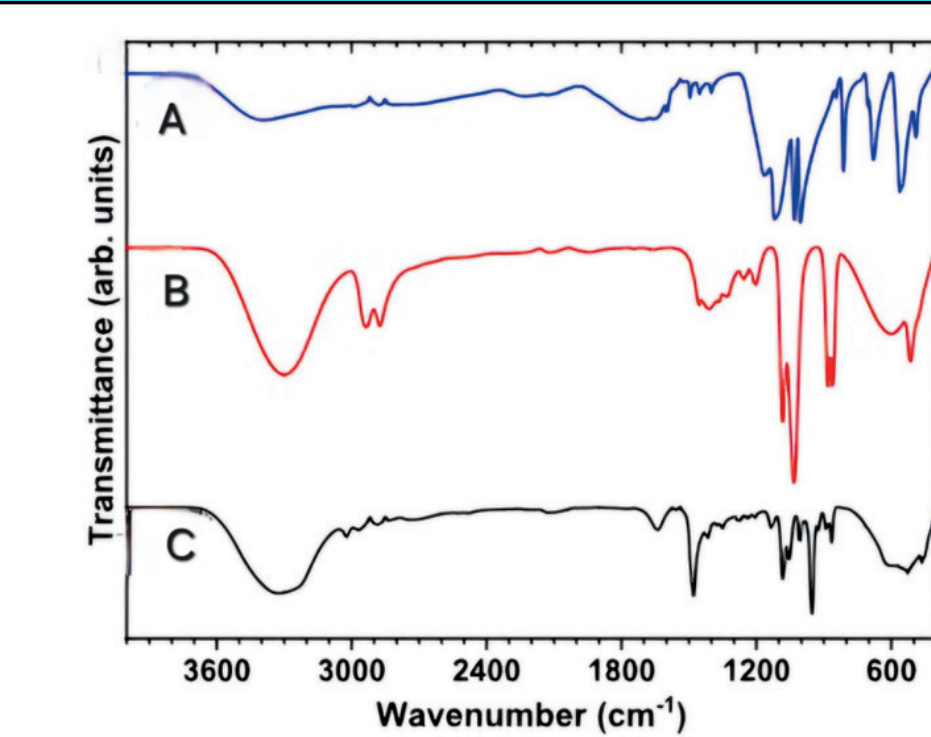


Fig 7: FTIR analysis for components A, B & C

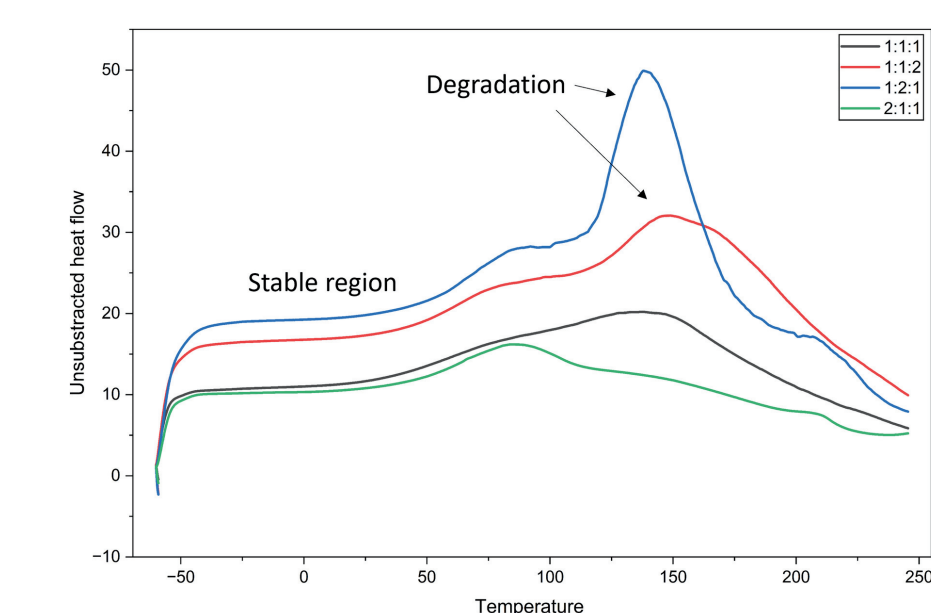


Fig 8: Behaviour of TDES to heat flow

Metal	Conc. (ppm)	Weight Fraction	Conc. (ppm)	Weight Fraction	Conc. (ppm)	Weight Fraction
Li	53.2	0.07092	155.331	0.07766	83.66	10.5968
Co	495.4	0.60213	1299.93	0.64996	79.7	10.0777
Mn	1.2	0.0027	89.829	0.04491	0.37	0.04649
Ni	<1 ppb	0	40.995	0.02049	619.333	78.3583

Table 1: Composition of Synthetic LCO and real cathode (from 30 batteries) obtained from ICP-MS results

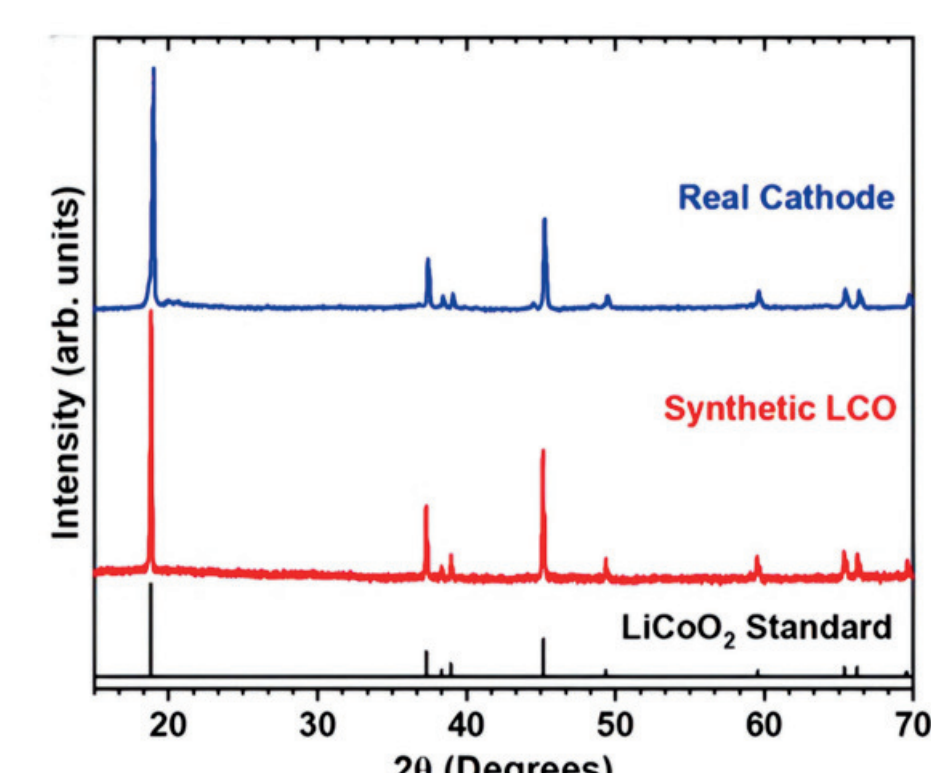


Fig 9: XRD analysis Real cathode and Synthetic LCO

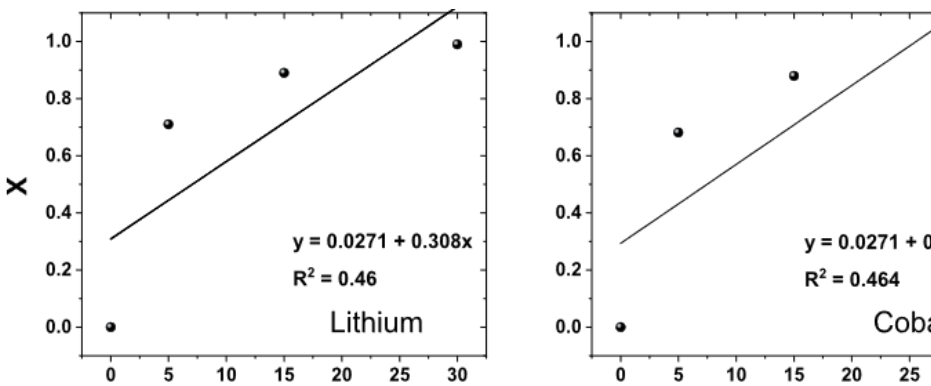


Fig 18: Modeling of the Synthetic LCO leaching data for liquid film mass transfer control

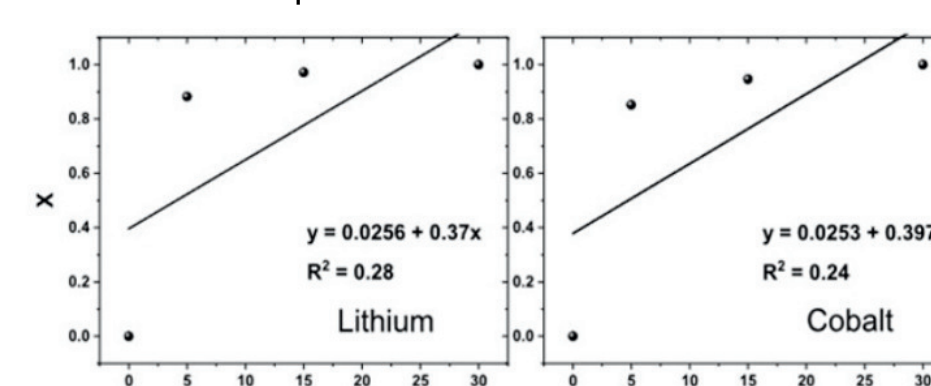


Fig 19: Modeling of the Synthetic LCO leaching data for product layer mass transfer control

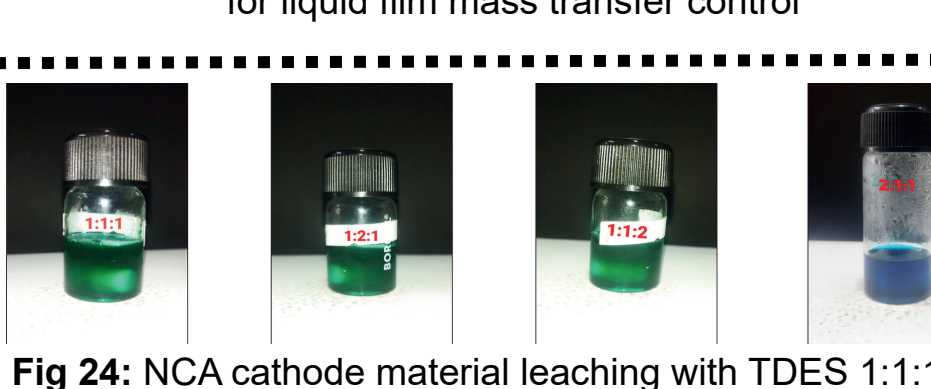


Fig 20: Modeling of the Synthetic LCO leaching data for chemical reaction control

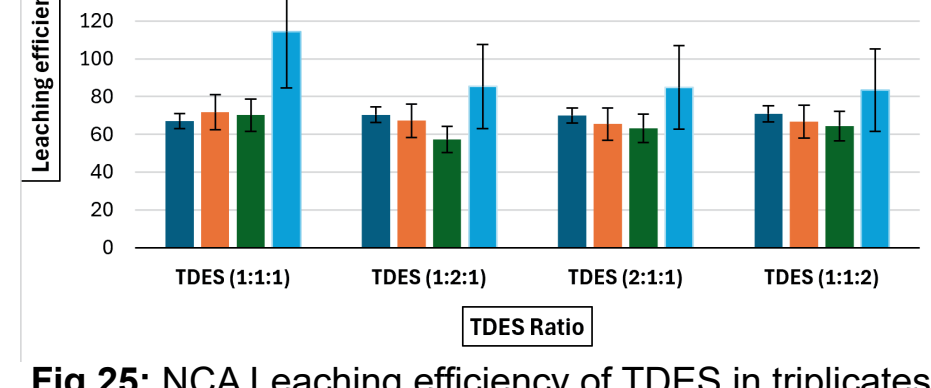


Fig 21: Modeling of the Real cathode leaching data for liquid film mass transfer control



Fig 22: Modeling of the Real cathode leaching data for product layer mass transfer control

Fig 23: Modeling of the Real cathode leaching data for chemical reaction control

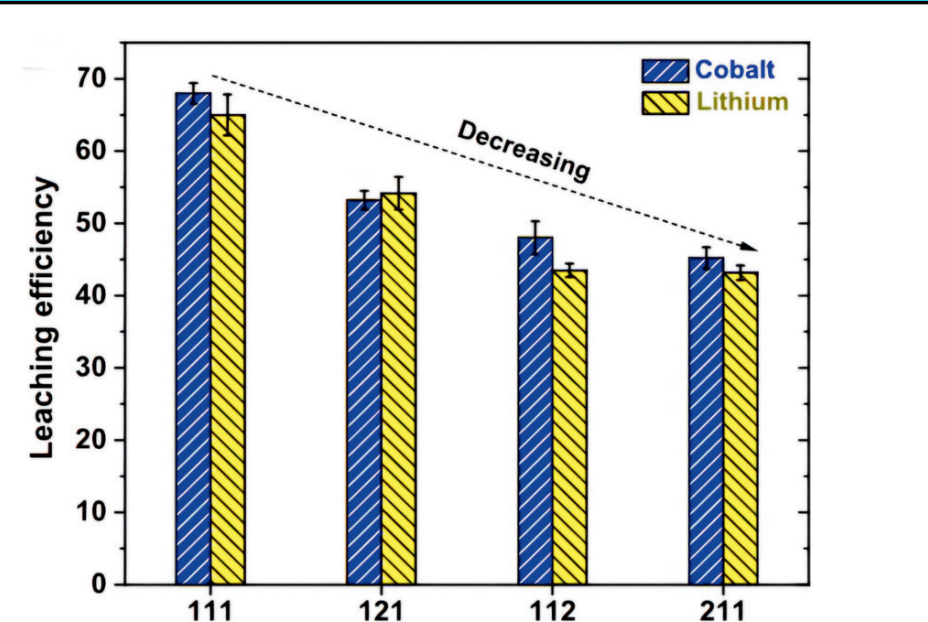


Fig 10: Leaching efficiency with Synthetic LCO

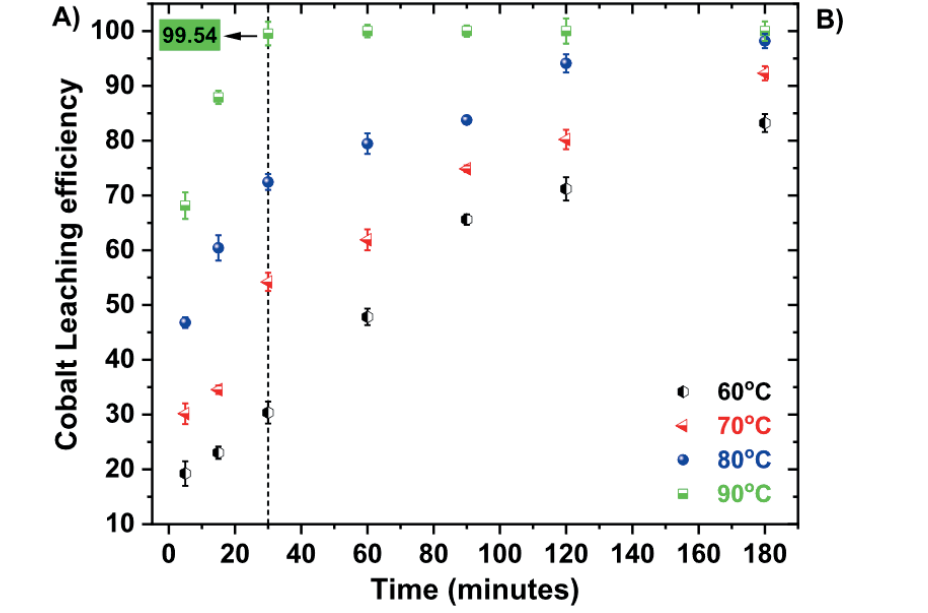


Fig 11: Leaching efficiency with Real cathode

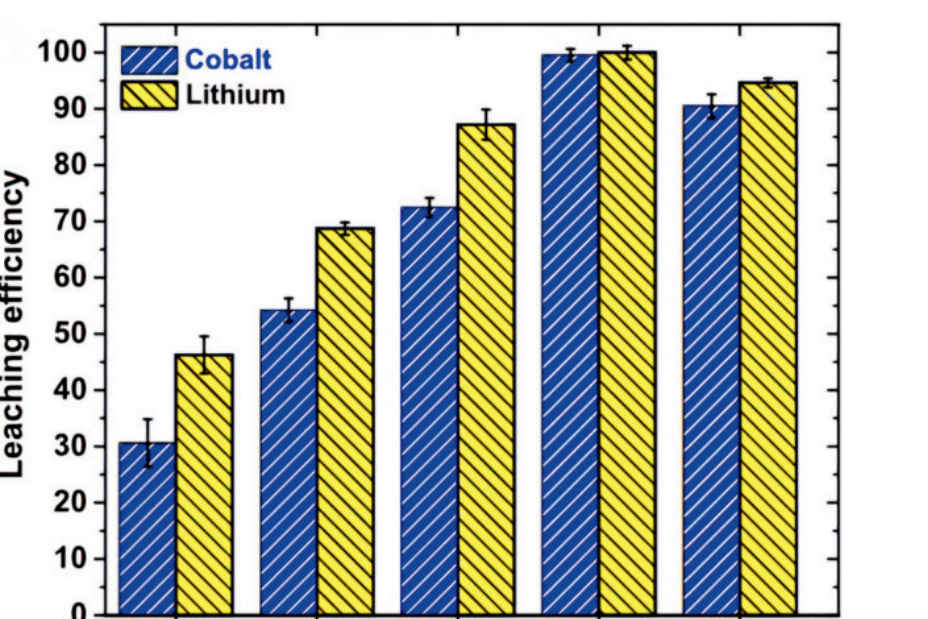


Fig 12: Leaching efficiencies of Cobalt with different temperatures at 20 g/L, 50 °C and 3h: Synthetic LCO

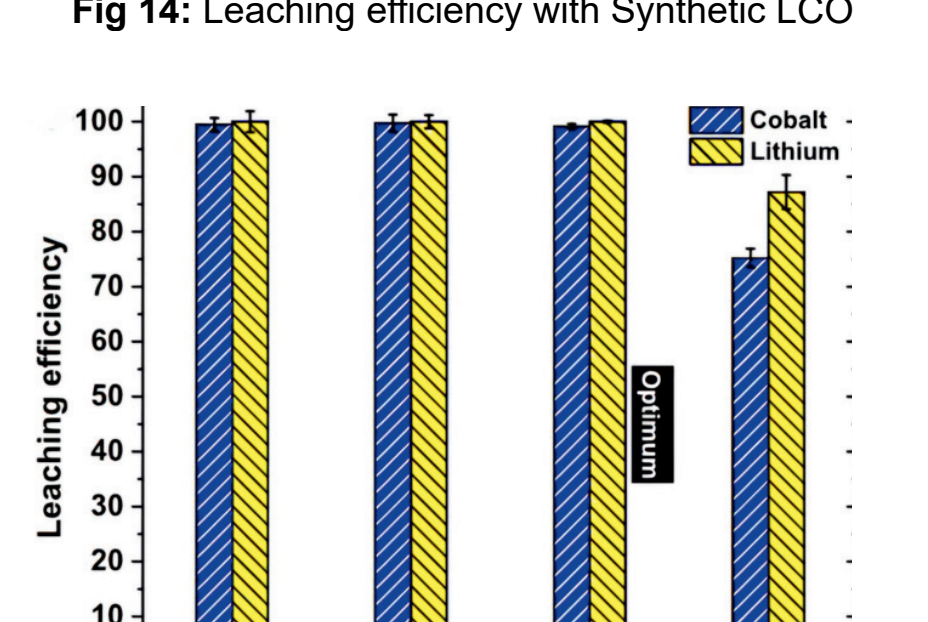


Fig 13: Leaching efficiencies of Cobalt with different temperatures at 20 g/L, 50 °C and 3h: Real cathode

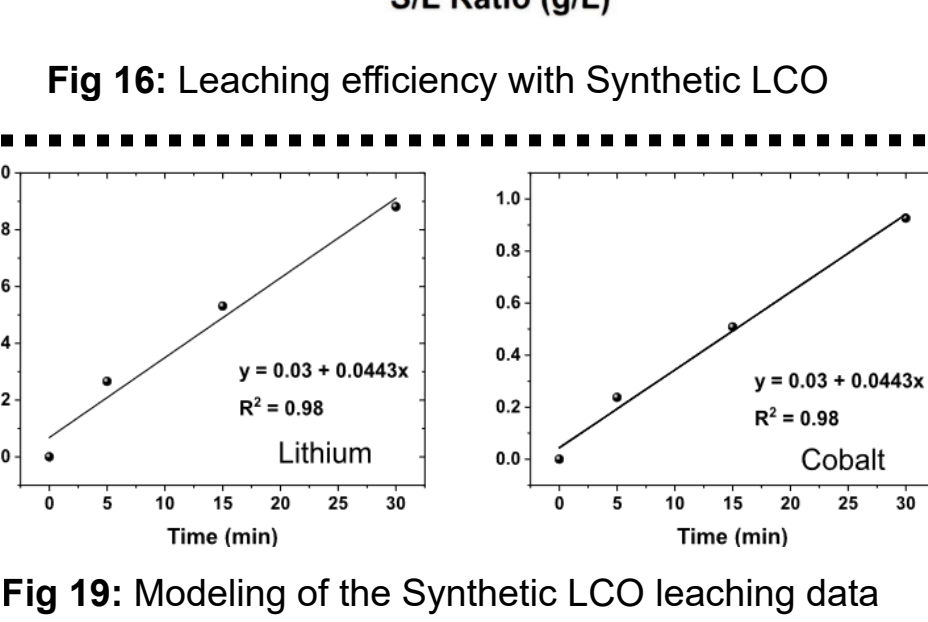


Fig 14: Leaching efficiency with Synthetic LCO

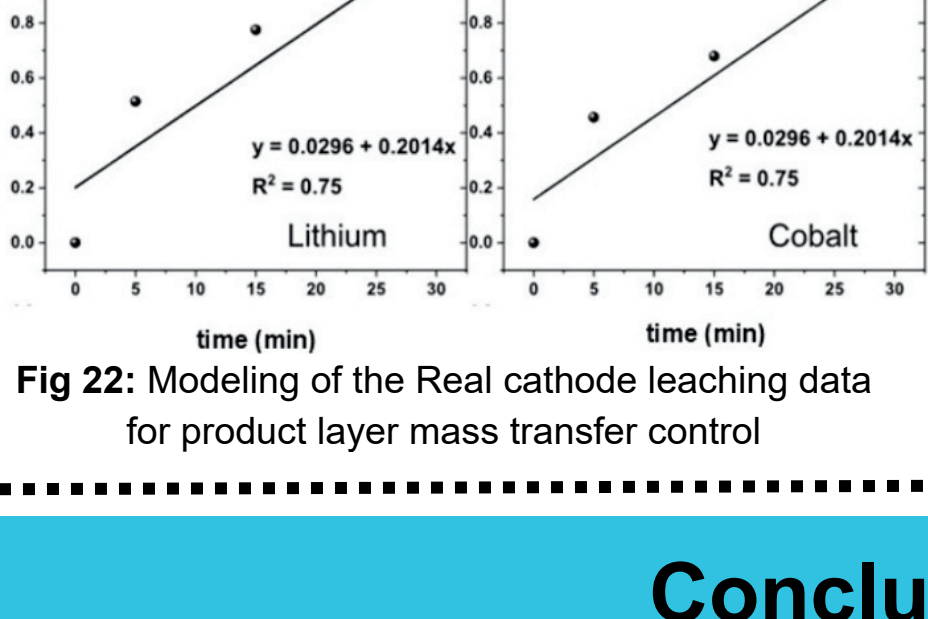


Fig 15: Leaching efficiency with real cathode

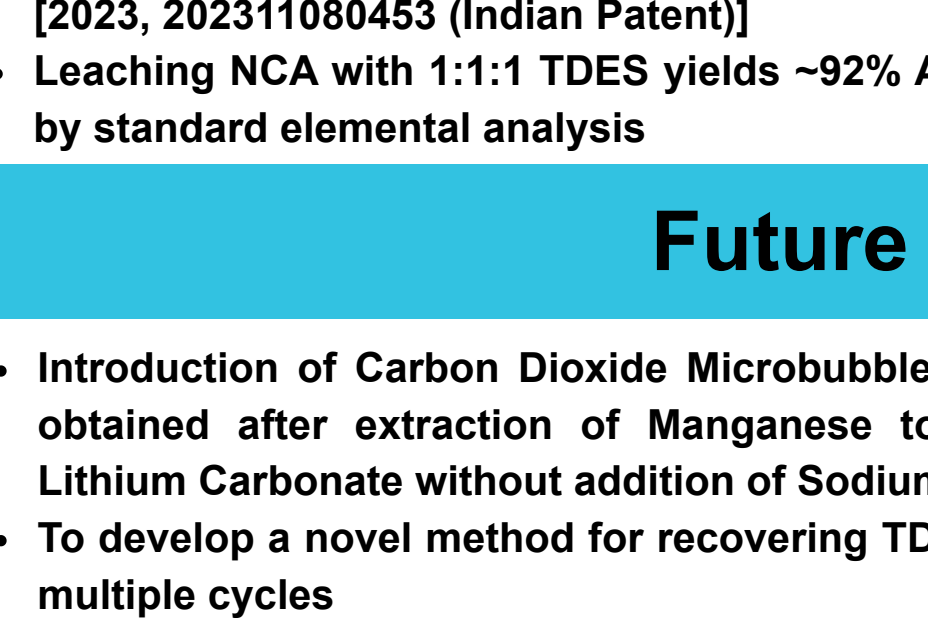


Fig 16: Leaching efficiency with Synthetic LCO

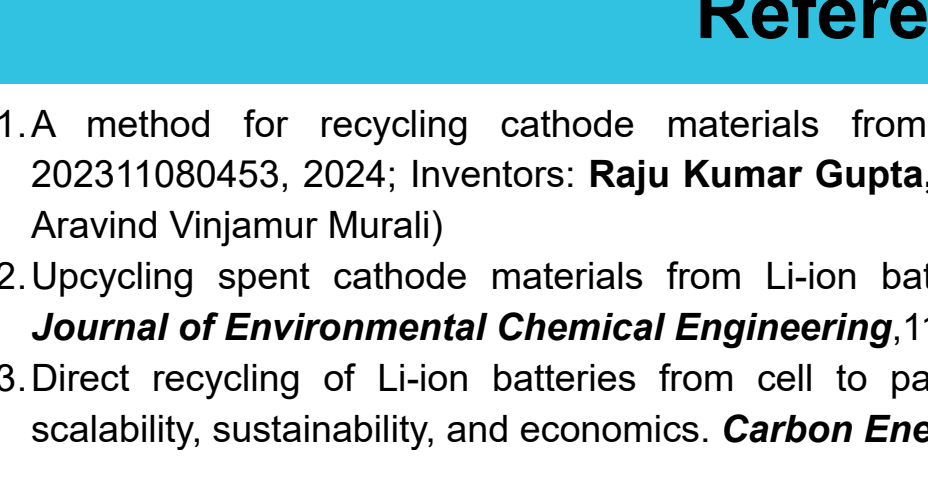


Fig 17: Leaching efficiency with Real cathode

Conclusions

- Optimum Leaching: 1:1:1 TDES, 20 g/L, 90 °C, 30 min => > 99% Co and > 75% Li efficiency. [2023, 202311080453 (Indian Patent)]
- Leaching NCA with 1:1:1 TDES yields ~92% Al, ~70% Li, ~68% Ni, and ~64% Co, validated by standard elemental analysis

Future Work

- Introduction of Carbon Dioxide Microbubbles to the supernatant obtained after extraction of Manganese to obtain high purity Lithium Carbonate without addition of Sodium salts
- To develop a novel method for recovering TDES and reusing it for multiple cycles

Fig 26: Li2CO3 extraction using CO2 bubbling process

Acknowledgement



References

- A method for recycling cathode materials from Spent Lithium-ion batteries. (Indian Patent No: 202311080453, 2024; Inventors: Raju Kumar Gupta, Sudhir Ranganath, Hemanth Peddavenkatappagari, Aravind Vinjamur Murali)
- Upcycling spent cathode materials from Li-ion batteries to precursors: Challenges and opportunities. *Journal of Environmental Chemical Engineering*, 11(4), 112016, 2023.
- Direct recycling of Li-ion batteries from cell to pack level: Challenges and prospects on technology, scalability, sustainability, and economics. *Carbon Energy*, e492, 2024.